



Optimal stock investment strategy using prediction models

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- Models
- Investment Strategies
- Conclusion

Introduction

- Percentage of American adults who have invested in the stock market reached 61% (Jeffrey, 2023)
- As interest in the stock market continues to rise, the use of predictive models for stock investment analysis has emerged as a crucial research topic
- Predictive models serve as tools to forecast market directions and provide valuable information for making investment decisions
- This study analyzes firm and macroeconomic data, applying machine learning algorithms to select the best-performing model for predicting returns
- Based on these predictions, a portfolio is constructed with the highest-return companies to identify the optimal investment strategy

Data Description

- Firm Specific data : provides characteristics of individual companies
- Macroeconomic data : reflects the overall market conditions
- The data used in the analysis spans 25 years, from January 1997 to December 2021
- All data will be aggregated on a monthly basis
 - observed periods for the collected data vary, including Monthly, Quarterly, Semiannual, and Annual frequencies
 - to align all variables to a monthly frequency, we used the most recent value for variables that are not available monthly
- All variables were standardized to enhance the training speed and performance of the model

Data Description

Macroeconomic data

- 113 variables, collected from the FRED-MD database (McCracken and Ng, 2016)
- categorized into eight groups

Output and Income	evaluate the overall productivity and income levels of the economy	Money and Credit	money supply and credit conditions
Labor Market	employment-related indicators	Interest and Exchange Rates	interest rate and exchange rate indicators
Housing	housing indicators	Prices	price indicators, such as inflation and deflation measures
Consumption, Orders, and Inventories	indicators related to consumer spending, new orders, and business inventory levels	Stock Market	stock market indicators

Data Description

Fred	Description
RPI	Real Personal Income
W875RX1	Real personal income excluding current transfer receipts
INDPRO	Industrial Production: Total Index
IPFPNSS	Industrial Production: Final Products and Nonindustrial Supplies
IPFINAL	Industrial Production: Final Products
IPCONGD	Industrial Production: Consumer Goods
IPDCONGD	Industrial Production: Durable Consumer Goods
IPNCONGD	Industrial Production: Non-Durable Consumer Goods
IPBUSEQ	Industrial Production: Equipment: Business Equipment
IPMAT	Industrial Production: Materials
IPDMAT	Industrial Production: Durable Goods Materials
IPNMAT	Industrial Production: Non-Durable Goods Materials
IPMANSICS	Industrial Production: Manufacturing
IPB51222S	Industrial Production: Non-Durable Consumer Energy Products: Residential Utilities
IPFUELS	Industrial Production: Non-Durable Consumer Energy Products: Fuels
CUMFNS	Capacity Utilization: Manufacturing
CLF16OV	Civilian Labor Force Level
CE16OV	Employment Level
UNRATE	Unemployment Rate
UEMPMEAN	Average Weeks Unemployed
UEMPLT5	Number Unemployed for Less Than 5 Weeks
UEMP5TO14	Number Unemployed for 5-14 Weeks
UEMP15OV	Number Unemployed for 15 Weeks and over
UEMP15T26	Number Unemployed for 15-26 Weeks
UEMP27OV	Number Unemployed for 27 Weeks and over
ICSA	Initial Claims
PAYEMS	All Employees, Total Nonfarm
USGOOD	All Employees, Goods-Producing
CES1021000001	All Employees, Mining, Quarrying, and Oil and Gas Extraction
USCONS	All Employees, Construction
MANEMP	All Employees, Manufacturing
DMANEMP	All Employees, Durable Goods
NDMANEMP	All Employees, Nondurable Goods
SRVPRD	All Employees, Service-Providing
USTPU	All Employees, Trade, Transportation, and Utilities
USWTRADE	All Employees, Wholesale Trade
USTRADE	All Employees, Retail Trade
USFIRE	All Employees, Financial Activities
USGOVT	All Employees, Government
CES0600000007	Average Weekly Hours of Production and Employees, Goods-Producing
AWOTMAN	Average Weekly Overtime Hours of Production and Employees, Manufacturing
AWHMAN	Average Weekly Hours of Production and Employees, Manufacturing
CES0600000008	Average Hourly Earnings of Production and Nonsupervisory Employees, Goods-Producing
CES2000000008	Average Hourly Earnings of Production and Nonsupervisory Employees, Construction
CES3000000008	Average Hourly Earnings of Production and Nonsupervisory Employees, Manufacturing
HOUST	Housing Units Started: Total
HOUSTNE	Housing Units Started: Northeast Census Region
HOUSTMW	Housing Units Started: Midwest Census Region
HOUSTS	Housing Units Started: South Census Region
HOUSTW	Housing Units Started: West Census Region

Fred	Description
PERMIT	Housing Units Authorized in Permit-Issuing Places: Total
PERMITNE	Housing Units Authorized in Permit-Issuing Places: Northeast Census Region
PERMITMW	Housing Units Authorized in Permit-Issuing Places: Midwest Census Region
PERMITS	Housing Units Authorized in Permit-Issuing Places: South Census Region
PERMITW	Housing Units Authorized in Permit-Issuing Places: West Census Region
DPCERA3M086SBEA	Real personal consumption expenditures
CMRMTSPL	Real Manufacturing and Trade Industries Sales
MRTSSM44X72USS	Retail Sales: Retail Trade and Food Services
DGORDER	Manufacturers' New Orders: Durable Goods
ACOGNO	Manufacturers' New Orders: Consumer Goods
ANDENO	Manufacturers' New Orders: Nondefense Capital Goods
AMDMUO	Manufacturers' Unfilled Orders: Durable Goods
BUSINV	Total Business Inventories
ISRATIO	Total Business: Inventories to Sales Ratio
M1SL	M1 Money Stock
M2SL	M2 Money Stock
M2REAL	Real M2 Money Stock
TOTRESNS	Total Reserves of Depository Institutions
NONBORRES	Reserves of Depository Institutions
BUSLOANS	Commercial and Industrial Loans, All Commercial Banks
REALLN	Real Estate Loans, All Commercial Banks
NONREVSL	Nonrevolving Consumer Credit Owned and Securitized
DTCOLNVHFNM	Consumer Motor Vehicle Loans Owned by Finance Companies, Level
DTCTHFNM	Total Consumer Loans and Leases Owned and Securitized by Finance Companies, Level
SBCACBW027SBOG	Securities in Bank Credit, All Commercial Banks
FEDFUNDS	Federal Funds Effective Rate
TB3MS	3-Month Treasury Bill Secondary Market Rate, Discount Basis
TB6MS	6-Month Treasury Bill Secondary Market Rate, Discount Basis
GS1	Market Yield on U.S. Treasury Securities at 1-Year Constant Maturity
GS5	Market Yield on U.S. Treasury Securities at 5-Year Constant Maturity
GS10	Market Yield on U.S. Treasury Securities at 10-Year Constant Maturity
AAA	Moody's Seasoned Aaa Corporate Bond Yield
BAA	Moody's Seasoned Baa Corporate Bond Yield
TB3SMFFM	3-Month Treasury Bill Minus Federal Funds Rate
TB6SMFFM	6-Month Treasury Bill Minus Federal Funds Rate
T1YFFM	1-Year Treasury Constant Maturity Minus Federal Funds Rate
T5YFFM	5-Year Treasury Constant Maturity Minus Federal Funds Rate
T10YFFM	10-Year Treasury Constant Maturity Minus Federal Funds Rate
AAAFFM	Moody's Seasoned Aaa Corporate Bond Minus Federal Funds Rate
BAAFFM	Moody's Seasoned Baa Corporate Bond Minus Federal Funds Rate
EXSZUS	Swiss Francs to U.S. Dollar Spot Exchange Rate
EXJPUS	Japanese Yen to U.S. Dollar Spot Exchange Rate
EXUSUK	U.S. Dollars to U.K. Pound Sterling Spot Exchange Rate
EXCAUS	Canadian Dollars to U.S. Dollar Spot Exchange Rate
WPSFD49207	Producer Price Index by Commodity: Final Demand: Finished Goods
WPSFD49502	Producer Price Index by Commodity: Final Demand: Finished Consumer Goods
WTISPLC	Spot Crude Oil Price: West Texas Intermediate (WTI)
PPICMM	Producer Price Index by Commodity: Metals and Metal Products: Primary Nonferrous Metals

Data Description

Firm Specific data

- 51 variables, obtained from the CRSP and Compustat database (Green et al., 2017)
- categorized into six groups

Investment	Examines a company's spending on growth and assets	Value-versus-growth	Compares undervalued stocks to high-growth stocks
Intangibles	Non-physical assets like brand and IP that add value	Momentum	Stocks that recently performed well may continue to do so
Profitability	Looks at how effectively a company generates profit	Frictions	Costs and limitations in trading, like fees and taxes

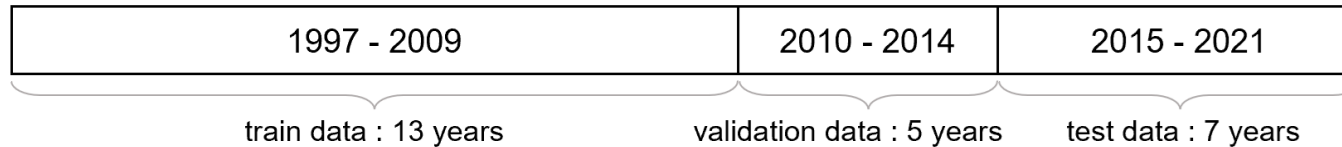
Data Description

Acronym	Description	Category
acc	Operating Accruals	Investment
adm	Advertising Expense-to-market	Intangibles
agr	Asset growth	Investment
alm	Quarterly Asset Liquidity	Intangibles
ato	Asset Turnover	Profitability
bm	Book-to-market equity	Value-versus-growth
bm_ia	Industry-adjusted book to market	Value-versus-growth
cash	Cash holdings	Value-versus-growth
cashdebt	Cash to debt	Value-versus-growth
cfp	Cashflow to price	Value-versus-growth
chcsho	Change in shares outstanding	Investment
chpm	Change in profit margin	Profitability
chtx	Change in tax expense	Momentum
cinvest	Corporate investment	Investment
depr	Depreciation / PP&E	Momentum
dolvol	Dollar trading volume	Frictions
dy	Dividend yield	Value-versus-growth
ep	Earnings-to-price	Value-versus-growth
gma	Gross profitability	Investment
grltnoa	Growth in long-term net operating assets	Investment
herf	Industry sales concentration	Intangibles
hire	Employee growth rate	Intangibles
ill	Illiquidity rolling (3 months)	Frictions
lev	Leverage	Value-versus-growth
lgr	Growth in long-term debt	Investment

Acronym	Description	
maxret	Maximum daily returns (3 months)	Frictions
me	Market equity	Frictions
me_ia	Industry-adjusted size	Frictions
mom1m	Previous month return	Momentum
mom6m	Cumulative Returns in the past (2-6) months	Momentum
mom12m	Cumulative Returns in the past (2-12) months	Momentum
mom36m	Cumulative Returns in the past (13-35) months	Momentum
mom60m	Cumulative Returns in the past (13-60) months	Momentum
ni	Net Equity Issue	Investment
nincr	Number of earnings increases	Momentum
noa	Net Operating Assets	Investment
op	Operating profitability	Profitability
pctacc	Percent operating accruals	Investment
pm	Profit margin	Profitability
pscore	Performance Score	Profitability
rd_sale	R&D to sales	Intangibles
rdm	R&D to market	Intangibles
rna	Return on Net Operating Assets	Profitability
roa	Return on Assets	Profitability
roe	Return on Equity	Profitability
rsup	Revenue surprise	Momentum
sgr	Sales growth	Value-versus-growth
sp	Sales-to-price	Value-versus-growth
std.dolvol	Std of dollar trading volume (3 months)	Frictions
std.turn	Std of Share turnover (3 months)	Frictions
turn	Shares turnover	Frictions

Data Description

- Companies with excessively low market capitalization were excluded
 - Data was collected annually over 25 years, targeting only the top 1,000 companies by market capitalization each year
 - The resulting dataset includes approximately 4,200 unique companies
- The monthly return can be calculated as follows : $r_{k,t} = \frac{S_{k,t} - S_{k,t-1}}{S_{k,t-1}}$
 - $r_{k,t}$: Return on stock k at time t
 - $S_{k,t}$: Price of stock k at time t
 - $S_{k,t-1}$: Price of stock k at time $t - 1$
- Combine the Firm Specific data and Macroeconomic data based on each month to create unified dataset



Models

1) Dimension Reduction

- Excessive number of explanatory variables can hinder model performance, so dimensionality reduction can help address these issues
- Macroeconomic data
 - time-series data
 - use Long Short-Term Memory method to extract hidden states (Chen et al., 2019)
- Firm Specific data
 - does not have continuous values for all companies
 - use principal component analysis method (Zhong and Enke, 2017)
- To find the best model, prediction models are built using both the full dataset and dimensionally reduced versions
 - Macroeconomic data : reduced to 4, 10, 20, and 50 components from the original 113 variables
 - Firm Specific data : reduced to 4, 10, and 20 components from the original 51 variables
 - resulting in a total of 20 different dataset combinations

Models

2) Machine Learning Models

- The machine learning methodologies used for prediction are CatBoost, XGBoost, and LightGBM
- For each set combination, a machine learning model is applied, and the R-Square value is calculated by iteratively training and validating the model on the train and validation datasets

train		validation
1997 - 2009		2010
1997 - 2010		2011
1997 - 2011		2012
1997 - 2012		2013
1997 - 2013		2014

- For each individual set and model, we identify the optimal tuning parameters that yield the best performance on the validation data

Models

2) Machine Learning Models

- R-squared values for the machine learning models

Firm Specific Data dimension	Macroeconomic Data dimension	Machine Learning Methodology		
		CatBoost	XGBoost	LightGBM
4	4	0.0403	0.0389	0.0253
	10	0.0280	0.0199	0.0230
	20	0.0344	0.0405	0.0027
	50	0.0336	0.0104	-0.0384
	113	0.0372	0.0227	0.0224
10	4	0.0354	0.0476	0.0249
	10	0.0269	0.0211	0.0206
	20	0.0366	0.0379	-0.0144
	50	0.0327	0.0109	-0.0747
	113	0.0385	0.0177	0.0223
20	4	0.0367	0.0388	0.0261
	10	0.0273	0.0211	0.0237
	20	0.0336	0.0369	-0.0283
	50	0.0295	0.0104	-0.0607
	113	0.0313	0.0145	0.0227
51	4	0.0439	0.0420	0.0231
	10	0.0311	0.0225	0.0205
	20	0.0309	0.0382	0.0004
	50	0.0294	0.0188	-0.0380
	113	0.0312	0.0084	-0.0221

- The current number of components used was arbitrarily selected, so we cannot be certain that these are the optimal results
- The three highlighted combinations and their corresponding models are set as the primary candidates
- The number of reduced dimensions is then adjusted incrementally, and model fitting is attempted again to identify the combination with the best performance

Models

2) Machine Learning Models

- R-squared values for the better machine learning models

Candidate 1		Macroeconomic data dimension				
Method : XGBoost		2	3	4	5	6
Firm Specific data dimension	8	0.0500	0.0293	0.0456	0.0251	0.0348
	9	0.0389	0.0285	0.0445	0.0252	0.0352
	10	0.0486	0.0329	0.0476	0.0249	0.0333
	11	0.0390	0.0287	0.0377	0.0265	0.0311
	12	0.0413	0.0283	0.0426	0.0249	0.0344
Candidate 2		Macroeconomic data dimension				
Method : CatBoost		2	3	4	5	6
Firm Specific data dimension	49	0.0507	0.0470	0.0365	0.0384	0.0418
	50	0.0533	0.0478	0.0393	0.0410	0.0382
	51	0.0516	0.0495	0.0439	0.0341	0.0437
Candidate 3		Macroeconomic data dimension				
Method : XGBoost		2	3	4	5	6
Firm Specific data dimension	49	0.0442	0.0309	0.0411	0.0237	0.0333
	50	0.0437	0.0318	0.0402	0.0236	0.0324
	51	0.0340	0.0446	0.0420	0.0228	0.0285

- Performance improvements were observed across all candidate models
- Using the three candidates that showed optimal performance, fitting is conducted on the test data

Models

2) Machine Learning Models

- Candidate models

Result	Methodology	Firm Specific data dimension	Macroeconomic data dimension	R-squared value
Candidate 1	XGBoost	8	2	0.0500
Candidate 2	CatBoost	50	2	0.0533
Candidate 3	XGBoost	51	3	0.0446

- Performance improvements were observed across all candidate models
- Using the three candidates that showed optimal performance, fitting is conducted on the test data

Models

3) Selections of the Portfolio

- For the test data, monthly returns for each company are predicted over the 7-year period from 2015 to 2021

train		test
1997 - 2014		2015
1997 - 2015		2016
⋮		
1997 - 2019		2020
1997 - 2020		2021

- R-squared values for the test data

Candidate 1	Candidate 2	Candidate 3
-0.0025	-0.0132	0.0071

→ The dataset and methodology of the third candidate with the best performance are selected as the final dataset and model

Investment Strategies

- When constructing an investment portfolio, two key decisions must be made:
 - which companies to invest in
 - how much to invest in each company
- If investments are made in n companies each month, the portfolio will be structured as follows:

$$P = w_1x_1 + \dots + w_nx_n$$

- where, P : current value of the portfolio
- $w_1, \dots, w_n$: weight ($\sum_{i=1}^n w_i = 1$)
- $x_1, \dots, x_n$: stocks (companies)

Investment Strategies

- The selection of companies to invest in each month is determined by two methods:
 1. Model Based : Using a return prediction model, select the top n companies based on their forecasted returns for the month
 2. Market Capitalization Based: Select the top n companies each month based on their market capitalization
- The investment strategy is determined using the following two methods:
 1. Equal Weight Investment : Allocate an equal amount of investment to each company
$$w_i = \frac{1}{n} \quad (i = 1, \dots, n)$$
 2. Capital Weighted Investment : Allocate investment funds in proportion to each company's market capitalization

$$w_i \propto \text{cap}(x_i)$$

Investment Strategies

1) Model based Investment vs Market Capitalization based Investment

- An investment portfolio was constructed by investing in 10 companies each month from January 2015 to December 2021
- The initial investment amount was uniformly set at \$10,000 for all comparisons
- Performance of the prediction model was evaluated by comparing the results of investments in Model based companies with those of investments in Market Capitalization based companies and the S&P 500 Index

Investment Strategies

1) Model based Investment vs Market Capitalization based Investment

- Annual returns for the Top10 Companies by Market Capitalization

Date	Equal Weight Investment returns	Capital Weighted Investment returns	Equal Weight Investment return rate	Capital Weighted Investment return rate
2015-12	10319.99	10409.57	3.20%	4.10%
2016-12	10857.53	10833.66	5.21%	4.07%
2017-12	14024.62	14416.59	29.17%	33.07%
2018-12	15114.26	15961.53	7.77%	10.72%
2019-12	17786.10	19509.37	17.68%	22.23%
2020-12	22993.54	27974.38	29.28%	43.39%
2021-12	32914.64	39329.08	43.15%	40.59%
Average	-	-	19.35%	22.60%

Market Capitalization Based
→ final amount of \$39,329

- Annual returns for the Top10 Companies by Model

Date	Equal Weight Investment returns	Capital Weighted Investment returns	Equal Weight Investment return rate	Capital Weighted Investment return rate
2015-12	9260.85	11145.88	-7.39%	11.46%
2016-12	19520.83	13714.30	110.79%	23.04%
2017-12	24676.16	16597.54	26.41%	21.02%
2018-12	20143.74	16007.01	-18.37%	-3.56%
2019-12	20490.08	14868.16	1.72%	-7.11%
2020-12	22512.25	17214.71	9.87%	15.78%
2021-12	32405.99	30540.53	43.95%	77.41%
Average	-	-	23.85%	19.72%

Model Based
→ final amount of \$32,405

Investment Strategies

1) Model based Investment vs Market Capitalization based Investment

- Annual returns for the S&P 500 Index

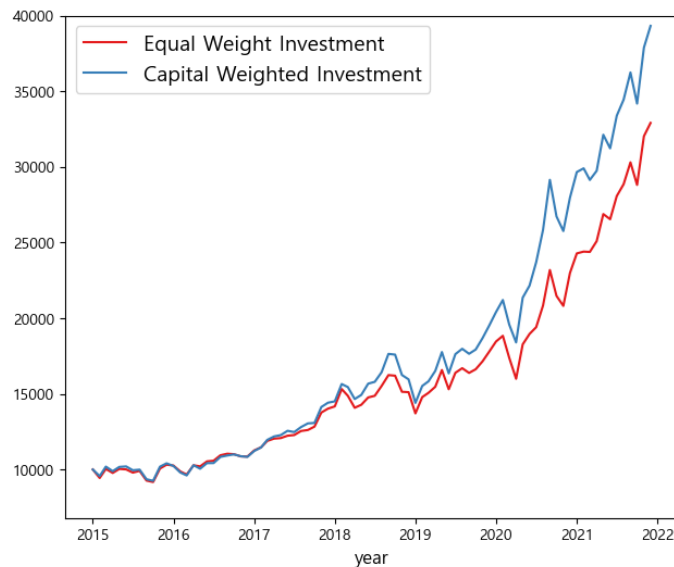
Date	Investment returns	Investment return rate
2015-12	10245.36	2.45%
2016-12	11222.26	9.54%
2017-12	13401.62	19.42%
2018-12	12565.73	-6.24%
2019-12	16194.47	28.88%
2020-12	18827.51	16.26%
2021-12	23890.75	26.89%
Average	-	13.89%

→ final amount of \$23,890

Investment Strategies

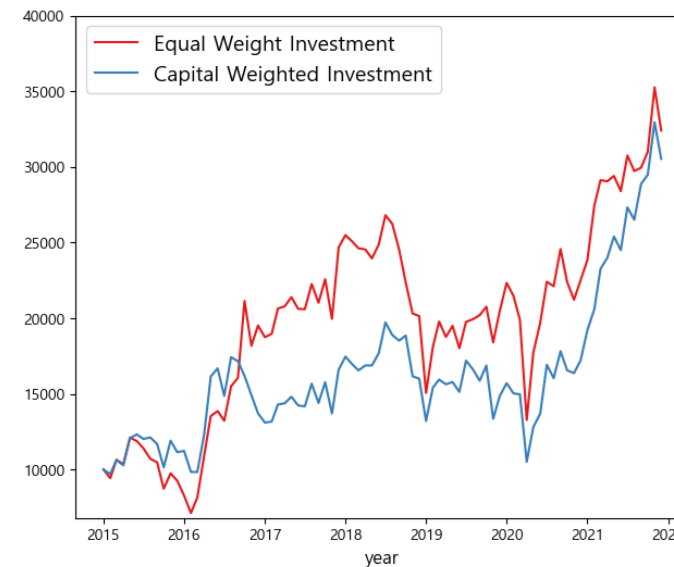
1) Model based Investment vs Market Capitalization based Investment

Market Capitalization Based strategy



→ Shows a steady upward trend

Model Based strategy



→ High volatility with a fluctuating pattern

Investment Strategies

2) Optimal Portfolio

- We modified the number of companies invested in each month and analyzed the investment returns
- Investments were made monthly in 1, 3, 5, 20, 30, 50, and 100 companies from 2015 to 2021

Data	top <i>n</i>	Equal Weight Investment Final Amount	Capital Weighted Investment Final Amount	Equal Weight Investment Average Return	Capital Weighted Investment Average Return	Equal Weight Investment SD of Returns	Capital Weighted Investment SD of Returns
Market Capitalization based	1	46204.58	46204.58	26.88%	26.88%	25.44%	25.44%
	3	63910.36	60704.41	31.78%	30.93%	19.60%	20.16%
	5	44657.99	48642.81	24.87%	26.61%	16.27%	18.01%
	10	32914.64	39329.08	19.35%	22.60%	13.93%	15.55%
	20	26253.41	32795.00	15.07%	19.01%	8.12%	11.20%
	30	26035.60	31661.98	14.81%	18.27%	6.14%	9.34%
	50	25130.34	29672.48	14.31%	17.20%	7.43%	9.61%
	100	22996.76	27466.15	12.90%	15.90%	7.79%	9.24%
Model based	1	8175.87	8175.87	160.95%	160.95%	426.37%	426.37%
	3	15219.06	50287.84	23.19%	44.85%	66.21%	75.75%
	5	30305.08	69790.91	24.96%	34.68%	47.91%	27.82%
	10	32405.99	30540.53	23.85%	19.72%	40.38%	25.87%
	20	25208.99	27150.40	16.16%	16.76%	22.27%	18.65%
	30	23101.52	29851.02	13.80%	17.72%	15.99%	13.95%
	50	23126.76	31240.41	13.52%	18.43%	13.56%	13.67%
	100	25642.19	26167.31	15.18%	15.44%	13.38%	12.86%

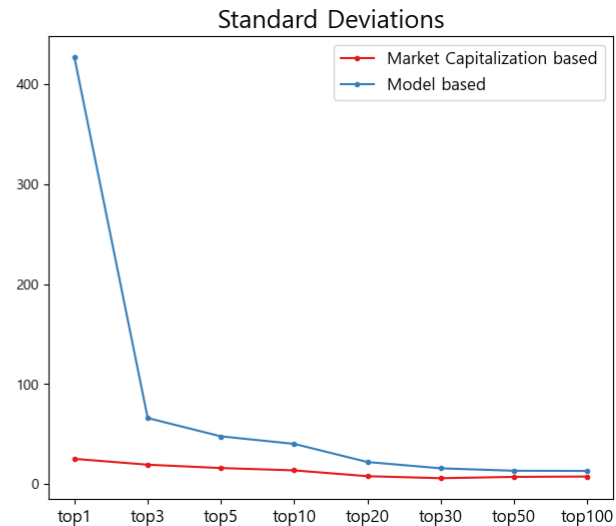
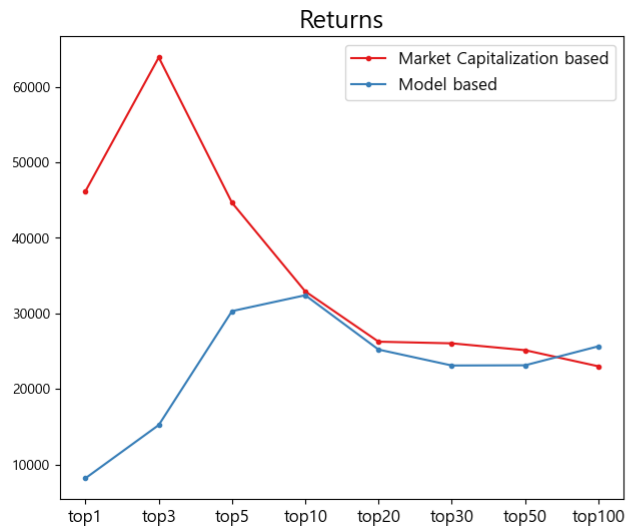
Market Capitalization Based
→ final amount of \$63,910

Model Based
→ final amount of \$69,790 (Maximum return)

Investment Strategies

2) Optimal Portfolio

- Equal Weight Investment

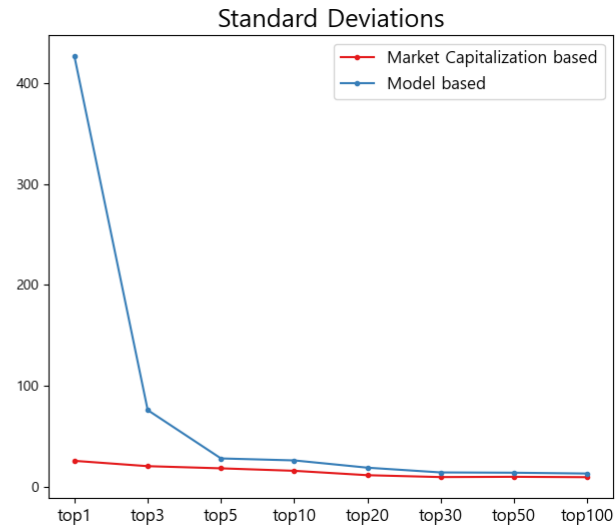
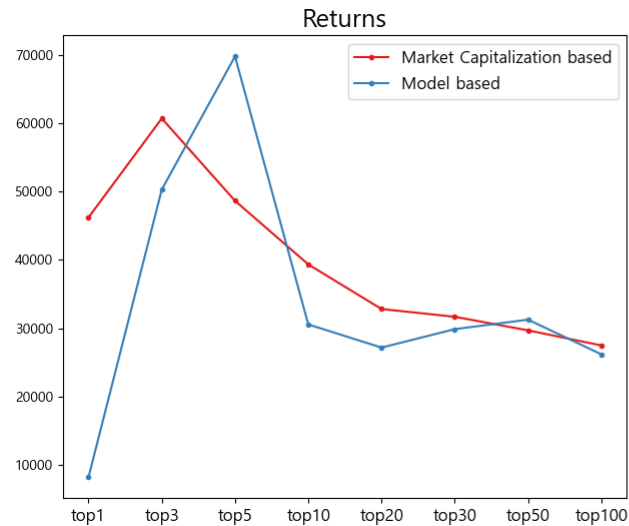


- Overall, investments based on market capitalization tended to yield higher final returns
- When selecting companies based on market capitalization, the annual return standard deviation was generally lower, showing a more stable pattern
- When selecting companies using the predictive model, volatility reached as high as 400%

Investment Strategies

2) Optimal Portfolio

- Capital Weighted Investment



- Overall, investments based on predictive model showed slightly higher final returns
- When selecting companies based on market capitalization, the annual return standard deviation was generally lower, showing a more stable pattern
- When selecting companies using the predictive model, volatility reached as high as 400%

Investment Strategies

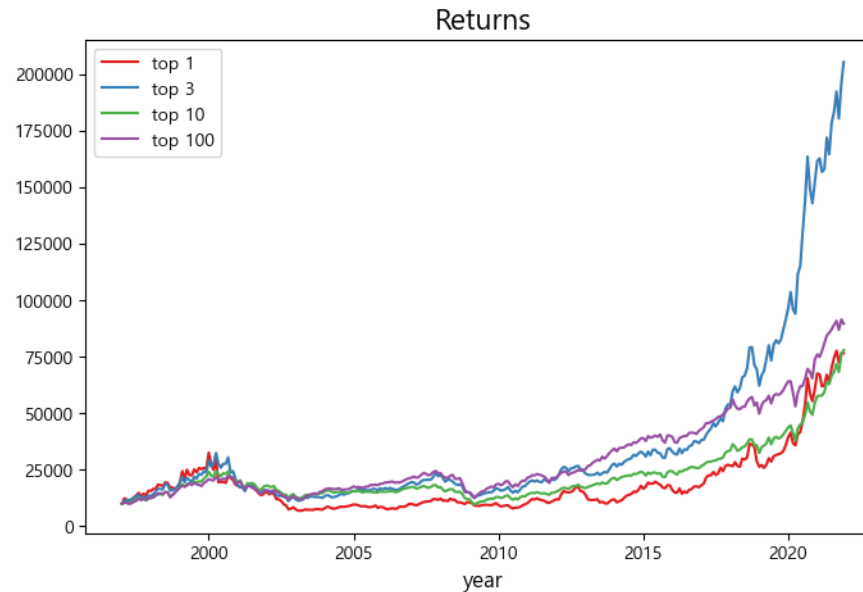
3) Long Term Results for Market Capitalization Weighted Investment

- When investing in Market Capitalization based companies, the previously calculated average annual returns were exceptionally high, reaching up to 30%
- The period from 2015 to 2021 was characterized by significant growth in the U.S. stock market, which may naturally account for these high returns
- Therefore, we extended the investment period to 1997-2021 to analyze long-term investments
- Investing in three companies per month resulted in the highest final amount

Data	top <i>n</i>	Equal Weight Investment Final Amount	Capital Weighted Investment Final Amount	Equal Weight Investment Average Return	Capital Weighted Investment Average Return	Equal Weight Investment SD of Returns	Capital Weighted Investment SD of Returns
Market Capitalization based	1	76495.97	76495.97	12.26%	12.26%	28.86%	28.86%
	3	205294.49	189251.76	15.07%	14.76%	21.91%	22.30%
	5	102493.32	116881.17	11.20%	11.98%	17.57%	18.93%
	10	77878.61	95571.05	9.93%	10.97%	16.93%	17.95%
	20	84077.43	95911.31	10.13%	10.85%	15.58%	16.67%
	30	91909.38	98398.47	10.37%	10.83%	14.68%	15.81%
	50	104769.52	101913.88	11.02%	10.99%	15.01%	15.77%
	100	89541.47	96040.86	10.42%	10.74%	15.39%	15.64%

Investment Strategies

3) Long Term Results for Market Capitalization Weighted Investment



- Until the early 2010s, the performance differences among the four investment strategies were not significant
- From the late 2010s, the strategy of investing in the top 3 market capitalization stocks exhibited a sharp increase in returns
- This suggests that the top 3 large-cap stocks played a leading role in the market

Investment Strategies

3) Long Term Results for Market Capitalization Weighted Investment

- Analyzed investments in Market Capitalization based companies during the period from 1997 to 2014, when the stock market did not experience significant growth
- Investing in a larger number of companies, specifically more than 50, resulted in the highest final investment amount

Data	top <i>n</i>	Equal Weight Investment Final Amount	Capital Weighted Investment Final Amount	Equal Weight Investment Average Return	Capital Weighted Investment Average Return	Equal Weight Investment SD of Returns	Capital Weighted Investment SD of Returns
Market Capitalization based	1	17838.32	17838.32	7.02%	7.02%	28.08%	28.08%
	3	32993.71	32343.99	8.73%	8.68%	19.12%	19.74%
	5	23452.62	24771.45	6.00%	6.46%	14.97%	15.97%
	10	24108.34	24910.54	6.38%	6.59%	16.55%	16.72%
	20	32385.16	29748.90	8.28%	7.77%	17.25%	17.34%
	30	35775.47	31627.28	8.73%	8.05%	16.55%	16.83%
	50	41910.76	34764.28	9.77%	8.65%	16.89%	16.97%
	100	38893.14	35205.79	9.45%	8.77%	17.38%	17.08%

Conclusion

- Investments in Market Capitalization based companies showed stable returns and low volatility
- Investments in Model based companies showed higher returns but with increased volatility
- Conservative investors are advised to invest in Market Capitalization based companies, while aggressive investors may prefer investing in Model based companies
- During extended periods of stock market growth, investing in fewer than 10 companies each month led to higher returns
- During periods without significant market growth, investing in more than 50 companies per month was preferable and provided more stable returns
- Future research could enhance the accuracy and reliability of stock return predictions by considering a wider array of variables and machine learning algorithms

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